App Risk Analyzer Report

App Name: Calculator

Package: com.miui.calculator

Version: 15.3.3

Risk Score: 10/100

Verdict: Low Risk

Permissions

Modify System Settings - Low

Widget Write Settings - Low

Network Status - Low

Wi-Fi - Low

System Permission Declare - Low

Internet - Low

Display Over Other Apps - Low

Widget Read Settings - Low

Multi-user Access - Low

Overlay Window - Low

Vibration - Low

Access Provider - Low

Update - Low

Location - Medium

Running Apps - Low

Foreground Service - Low

Access Note - Low

AI Security Analysis

■ Overall Security Analysis
The app “Calculator” requests a large set of system■level permissions that are unusual for a simple calculator. While the risk score (10/100) is low, the presence of high■privilege permissions indicates the app could be abused if compromised. Most of the functionality can be implemented without access to secure settings, overlay windows, or user■wide data.

■ Dangerous Permissions
- `android.permission.WRITE_SECURE_SETTINGS` – lets the app change system security settings (e.g., lock screen, device admin).
- `android.permission.SYSTEM_ALERT_WINDOW` / `android.permission.SYSTEM_OVERLAY_WINDOW` – allows drawing over other apps, which can be used for phishing or click■jacking.
- `android.permission.INTERACT_ACROSS_USERS_FULL` – gives control over other user profiles on the device.
- `android.permission.GET_TASKS` – reveals running apps and can be used for surveillance.

- `android.permission.FOREGROUND_SERVICE` – lets the app run continuously in the background, increasing persistence.
- `com.miui.securitycenter.permission.SYSTEM_PERMISSION_DECLARE` & other MIUI■specific widget permissions – grant deep integration with the MIUI system UI and settings.

■ Privacy Concerns
- `android.permission.ACCESS_COARSE_LOCATION` – can infer the user's approximate location.
- `android.permission.ACCESS_NETWORK_STATE` & `android.permission.INTERNET` – enable the app to monitor network usage and send data off■device.
- `com.xiaomi.market.sdk.UPDATE` – may allow silent app updates without user consent.
- `com.miui.notes.permission.ACCESS_NOTE` – could read personal notes stored by the MIUI Notes app.

■ Security Recommendations
- **Remove unnecessary permissions**: The calculator should not need secure settings, overlay windows, or cross■user access.
- **Implement least■privilege principle**: Request only `INTERNET` (if needed) and basic vibration.
- **Conduct code review**: Verify that any use of high■risk APIs is justified and properly sandboxed.
- **Monitor network traffic**